Concretely, a probability distribution over our action set A is denoted $P(A)$: $A_i \rightarrow [0,1]$, meaning that $P(A)$ is a map from a set A to a set of real numbers between 0 and 1. In particular, each element $a_i \in A$ is mapped to a single number between 0 and 1 such that the sum of all these numbers for each action is equal to 1. We might represent this map for our Gridworld action set as just a vector, where we identify each position in the vector with an element in the action set, e.g., [up, down, left, right] → [0.25, 0.25, 0.10, 0.4]. This map is called a *probability mass function* (PMF).

What we just described is actually a *discrete* probability distribution, since our action set was discrete (a finite number of elements). If our action set was infinite, i.e., a continuous variable like velocity, we would call this a *continuous* probability distribution and instead we would need to define a *probability density function* (PDF).

The most common example of a PDF is the normal (also known as Gaussian, or just bell-curve) distribution. If we have a probability with a continuous action, say a car game where we need to control the velocity of the car from 0 to some maximum value, which is a continuous variable, how might we do this with a policy network? Well, we could drop the idea of probability distribution and just train the network to produce the single value of velocity that it predicts is best, but then we'd risk not exploring enough (and it is difficult to train such a network). A lot of power comes from a little bit of randomness. The kind of neural networks we employ in this book only produce vectors (or tensors more generally) as output, so they can't produce a continuous probability distribution—we have to be more clever. A PDF like a normal distribution is defined by two parameters, the mean and variance. Once we have those, we have a normal distribution that we can sample from. So we can just train a neural network to produce mean and standard deviation values that we can then plug into the normal distribution equation and sample from that.

Don't worry if this isn't all making sense now. We will continue to go over it again and again because these concepts are ubiquitous in reinforcement learning and machine learning more broadly.